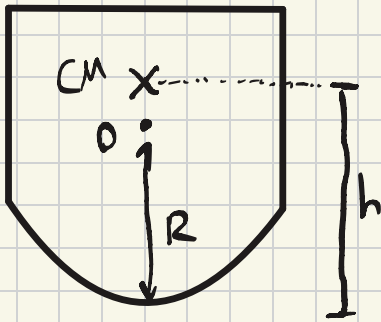


Ex 1: Kids Toy Stable Equilibrium



a) $h(\phi) = R - R \cos \phi + h \cos \phi$

$$= R + (h - R) \cos \phi$$

@ $\phi = 0 \rightarrow h(0) = R + (h - R) = h$ ✓

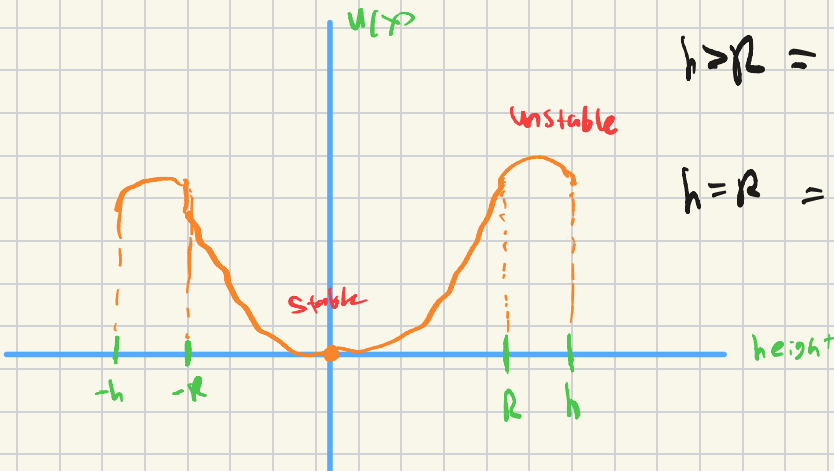
@ $\phi = \pi/2 \rightarrow h(\frac{\pi}{2}) = R$ ✓

b) $U = mg \cdot h \rightarrow mg(R + (h - R) \cos \phi)$

$h < R$ = Stable
(stays upright)

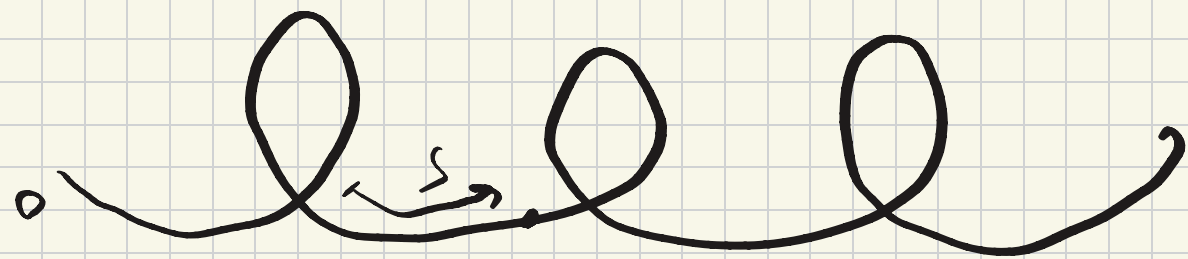
$h > R$ = Unstable
(falls over)

$h = R$ = Equal



c) The toy remains up right when $R < L$.
 This makes sense because when $R < L$, the
 $U(x)$ graph is in a Stable Equilibrium

Ex 2: 1D motion



a) $r(t) = x(t)\hat{i} + y(t)\hat{j}$ Cartesian Coords.

$$v = \dot{x}\hat{i} + \dot{y}\hat{j}$$

are NOT good for

$$|v| = \sqrt{\dot{x}^2 + \dot{y}^2}$$

this because x and y
 are not independent (constrained)

b) $v = \frac{ds}{dt} \hat{s}$
 $|v| = \dot{s}$

This is a better coordinate plane
 since speed only relies on the position
 on the wire and it is independent

$$c) F_{\text{net}} = m a$$

$$a_{||} = d^2 s / dt^2$$

$$F_{||} = m \cdot a_{||} \Rightarrow \boxed{m \cdot \ddot{s}}$$



$$d) F = - \frac{dU}{ds}$$

$$F_{||} = - \frac{dU}{ds}$$

$$m \ddot{s} = - \frac{dU}{ds}$$